

Units of mass



1 Choose the most appropriate unit of measure for the following:

- Grams
- Kilograms
- Tonnes
- Milligrams

- | | |
|----------------------------|------------------------|
| a Cup of flour | b Grain of sand |
| c Box of 50 oranges | d Biscuit |
| e Car | f Pencil case |
| g Your weight | h Aeroplane |

2 Convert the following to milligrams (mg):

- | | | | |
|--------------|------------------|-----------------|-------------------|
| a 6 g | b 0.006 g | c 0.47 g | d 72.668 g |
|--------------|------------------|-----------------|-------------------|

3 Convert the following to grams (g):

- | | | | |
|----------------|-----------------|------------------|--------------------|
| a 87 mg | b 155 mg | c 8505 mg | d 52 005 mg |
|----------------|-----------------|------------------|--------------------|

4 Convert the following to the indicated units:

- | | |
|--------------------------------|-------------------------------|
| a 6.251 kg to grams | b 12 kg into grams |
| c 946 g to kilograms | d 11 t into kilograms |
| e 3000 g into kilograms | f 5 t into kilograms |
| g 9.272 t to kilograms | h 0.493 t to kilograms |
| i 1000 kg into tonnes | j 8500 kg to tonnes |

5 Convert the following to the indicated units:

- | | |
|---------------------------------|-----------------------------------|
| a 7.538 kg to milligrams | b 0.75 t into grams |
| c 0.054 kg to milligrams | d 11 540 000 g into tonnes |

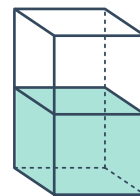
Estimation

6 State a reasonable estimate for the mass of each of the following:

- | | | | |
|-----------------------|--------------------|-------------------------|--------------------------|
| a Goat | b Cat | c 50 dollar bill | d Can of soup |
| e Standard die | f Newspaper | g Cheese | h Block of butter |



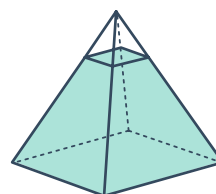
- 7 The container shown has 4 mg of jelly in it. Estimate the weight of the container when it is full.



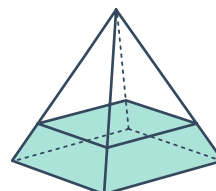
- 8 The container shown has 12 mg of water in it. Estimate the weight of the container when it is full.



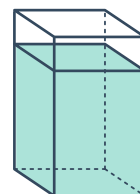
- 9 The container shown has 9 kg of liquid in it. Estimate the weight of the container when it is full.



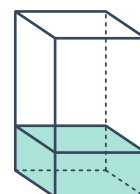
- 10 The container shown has 12 g of water in it. Estimate the weight of the container when it is full.



- 11 The container shown has 15 g of liquid in it. Estimate the weight of the container if it was full.



- 12 The container shown has 1 g of jelly in it. Estimate the weight of the container when it is full.

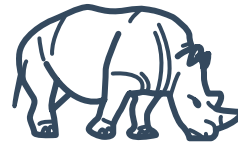
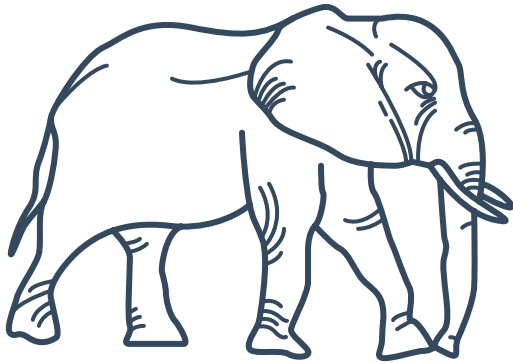


- 13 An elephant has a mass of 5000 kg:

Elephant



Rhinoceros



Use the diagram to estimate the mass of a rhinoceros.

- 14 A tablet has a mass of 640 grams:



Use the diagram to estimate the mass of a laptop.

Applications

- 15 Adam's toy truck has a mass of 200 g. Find the weight of the toy truck in milligrams.
- 16 Valerie has fruit everyday for dinner. Today she consumed 45 cherries, which weighed a total of 180 g. How many milligrams did the average cherry weigh?
- 17 Rewrite 2468 g in terms of the number of whole kilograms (kg) and grams (g) left over.

- 18 Jordan's bag weighs $\frac{1}{2}$ of a kilogram. How  grams does the bag weigh?
- 19 Find the total mass of 12 cans of soup, in kilograms, if each has a mass of 350 g.
- 20 The cost of mining gold for a mining company is \$840 per gram. How much will it cost the company to mine 0.2 kg of gold?
- 21 Justin is weighing himself and his dog at the same time. Their combined mass is 70 kg. Justin knows that he weighs 32 kg, so what must his dog weigh?
- 22 Mohamad's trampoline can hold a maximum of 90 kg of mass. If Mohamad has some friends over and they each weigh 30 kg, how many of them can use the trampoline at once?
- 23 At a sporting match, the coach has 10 oranges in a container to give out to the team members at half-time. The container of oranges weighs 800 g. How much does each orange weigh?
- 24 If 11 apples have a total mass of 3.19 kg, find the average mass of those apples in grams.
- 25 A truck is transporting 8 horses. Each horse in the truck weighs 800 kg.
- a How much do all the horses in the truck weigh altogether?
 - b If the truck weighed 4000 kg before the horses were loaded into it, how many kilograms does the loaded truck weigh now?
- 26 James buys a 4000 g box of washing powder. If he uses 2.3 kg of the washing powder, how many grams of powder is left in the box?
- 27 Christa buys a 8 kg bag of rice. If she uses 2800 g, how many kilograms of rice is left?
- 28 Passengers have a limit of 20 kg of luggage on a plane. If James's bag was over the weight limit by 1800 g, how much does his bag weigh in kilograms?
- 29 If each carrot has a mass of 70 g and each lemon has a mass of 60 g, how many kilograms would 6 carrots and 9 lemons weigh altogether?
- 30 If a truck can carry 720 kg of soil in one load, how many loads does it need to take in order to empty a field of 23.04 t of soil?
- 31 The nutritional information on a brand of corn chips states that there is 6 mg of sodium in every 100 g of corn chips. How many grams of sodium would there be in 10 packets of corn



32 If 25 bags of flour have a mass of 2 kg, find the mass of 5 bags, in grams.